Cambio tecnológico y empleo

- Existe un intenso debate en todo el mundo sobre el impacto del cambio tecnológico y la automatización de tareas en la naturaleza presente y futura del trabajo. La discusión, sin embargo, no es nueva. Durante la década de 1990 hubo consenso sobre el cambio tecnológico sesgado hacia mayores skills / habilidades (SBTC), especialmente en los países desarrollados.
- Esta fue la explicación canónica con respecto a la creciente demanda de trabajadores altamente calificados. Este proceso fue, a su vez, un factor que contribuyó al aumento de la desigualdad de ingresos (Katz y Autor,1999).
- Un nuevo fenómeno: los empleos de calificación media tuvieron una disminución respecto de las ocupaciones de alta y baja calificación / salarios bajos. Este fenómeno de polarización laboral se ha encontrado principalmente en los Estados Unidos (Wright y Dwyer, 2003; Autor y Dorn, 2013; Autor et al., 2006) y en algunos países europeos (Goos et al., 2014).
- Una explicación: automatización de las tareas rutinarias (cognitivas o manuales), generalmente realizadas por trabajadores de mediana calificación, por sobre tareas cognitivas o "manuales no rutinarias". Dado que las tareas rutinarias se encuentran en la parte media de la distribución salarial y las tareas no rutinarias en la parte superior e inferior, la tecnología genera una reducción de los salarios en el primer grupo y un aumento (o estabilidad) en el grupo (Goos y Manning, 2007; Autor y Dorn, 2013). Por lo tanto, la polarización de ocupaciones sería una explicación de la creciente desigualdad salarial (Acemoglu y Autor, 2011).
- La idea de Autor, Levy y Murnane (2003) del impacto de la tecnología en la demanda de diferentes habilidades permitiría explicar la polarización de las ocupaciones (Goos y Manning (2007)

THE SKILL CONTENT OF RECENT TECHNOLOGICAL CHANGE: AN EMPIRICAL EXPLORATION*

DAVID H. AUTOR
FRANK LEVY
RICHARD J. MURNANE

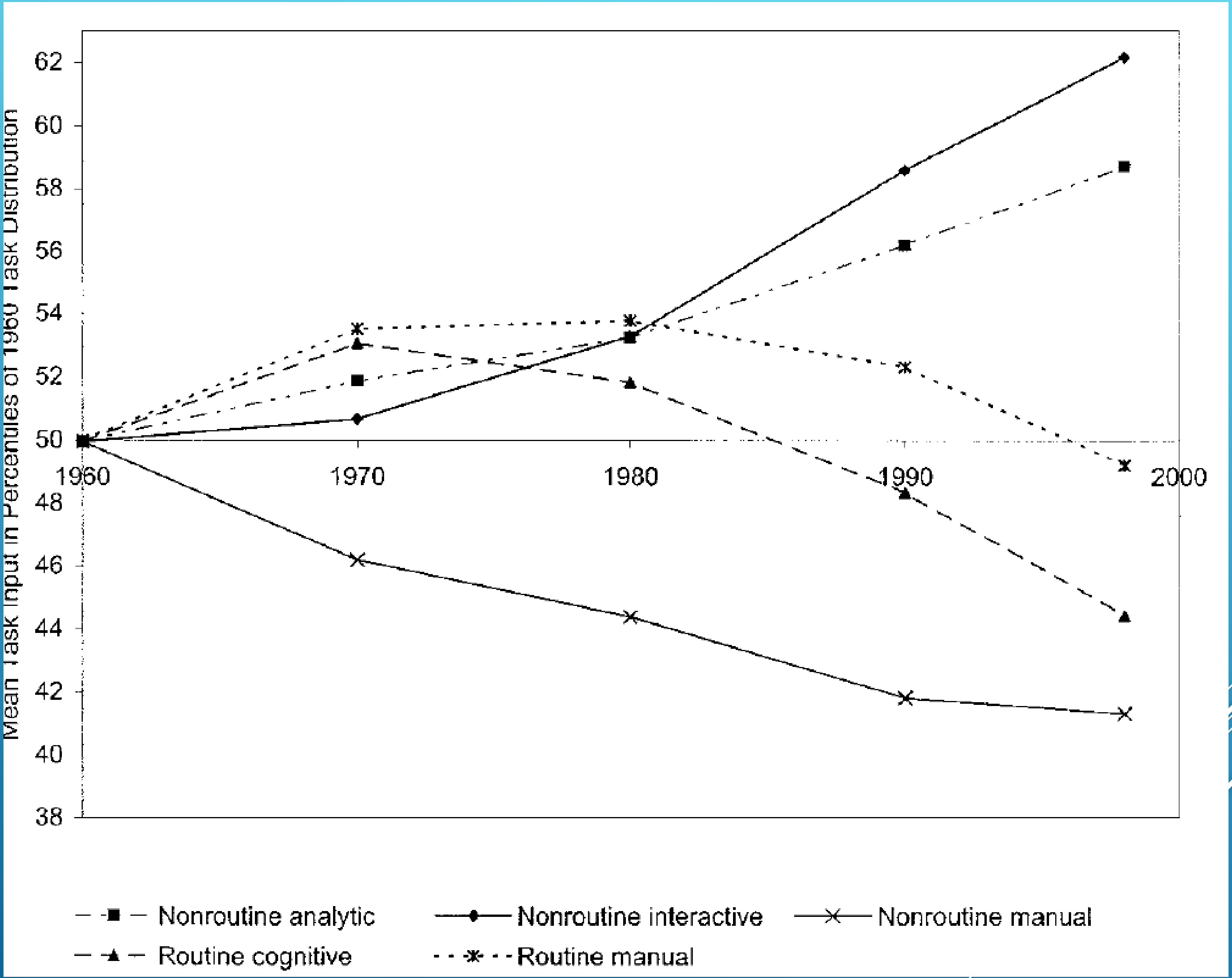
- **Argumento (1) tecnología sustituye a los trabajadores en la realización de tareas cognitivas y manuales que se pueden lograr siguiendo reglas explícitas y (2) complementaria de los trabajadores en la realización de tareas no rutinarias.**

- **Enfoque de tareas, más que de ocupaciones.**

TABLE I

PREDICTIONS OF TASK MODEL FOR THE IMPACT OF COMPUTERIZATION ON FOUR CATEGORIES OF WORKPLACE TASKS

	Routine tasks	Nonroutine tasks
	Analytic and interactive tasks	
Examples	• Record-keeping • Calculation • Repetitive customer service (e.g., bank teller)	• Forming/testing hypotheses • Medical diagnosis • Legal writing • Persuading/selling • Managing others
Computer impact	• Substantial substitution	• Strong complementarities
	Manual tasks	
Examples	• Picking or sorting • Repetitive assembly	• Janitorial services • Truck driving
Computer impact	• Substantial substitution	• Limited opportunities for substitution or complementarity

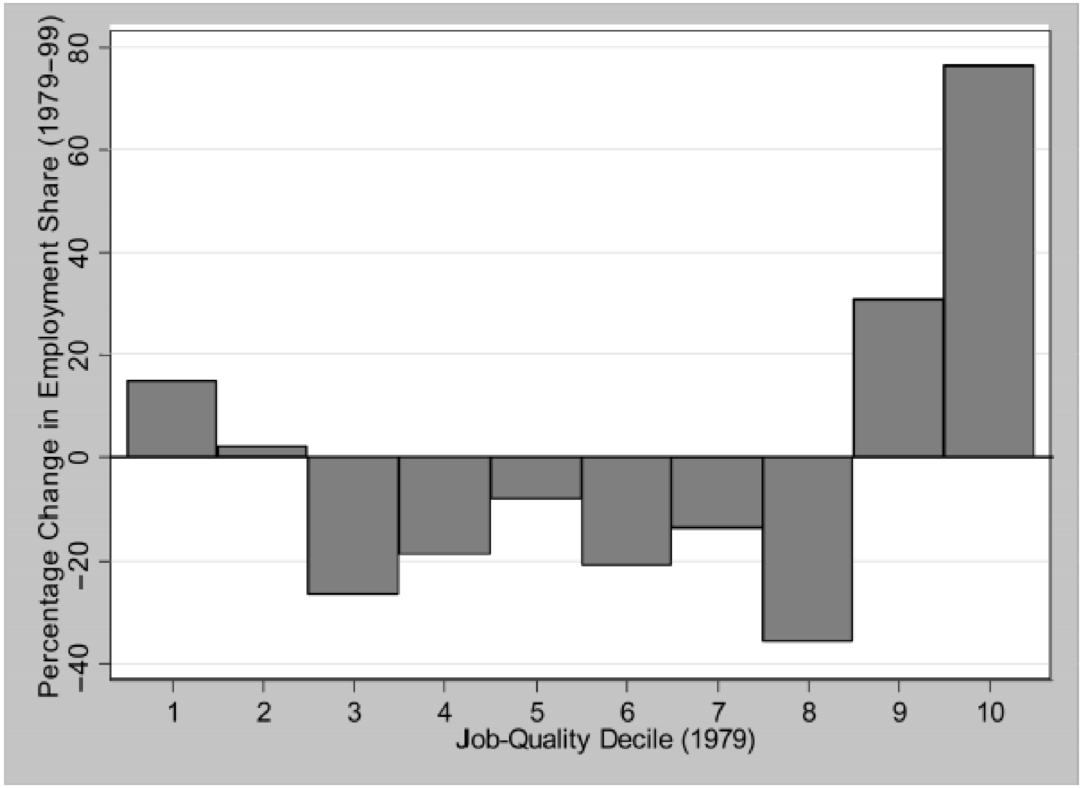


Mean Task Input in Percentiles of 1960 Task Distribution
62
60
58
56
54
52
50
48
46
44
42
40
38
1960
1970
1980
1990
2000
Nonroutine analytic
Nonroutine interactive
Nonroutine manual
Routine cognitive
Routine manual

LOUSY AND LOVELY JOBS: THE RISING POLARIZATION OF WORK IN BRITAIN

Maarten Goos and Alan Manning*

FIGURE 1.—PERCENTAGE CHANGE IN EMPLOYMENT SHARE BY JOB-QUALITY DECILE



Notes: Employment data are taken from the LFS using three-digit SOC90 codes. Employment changes are taken between 1979 and 1999. Quality deciles are based on three-digit SOC90 median wages in 1979 taken from the NES.

The models we estimate are of the quadratic form:

$$\Delta n_j = \beta_0 + \beta_1 w_{j0} + \beta_2 w_{j0}^2, \tag{1}$$

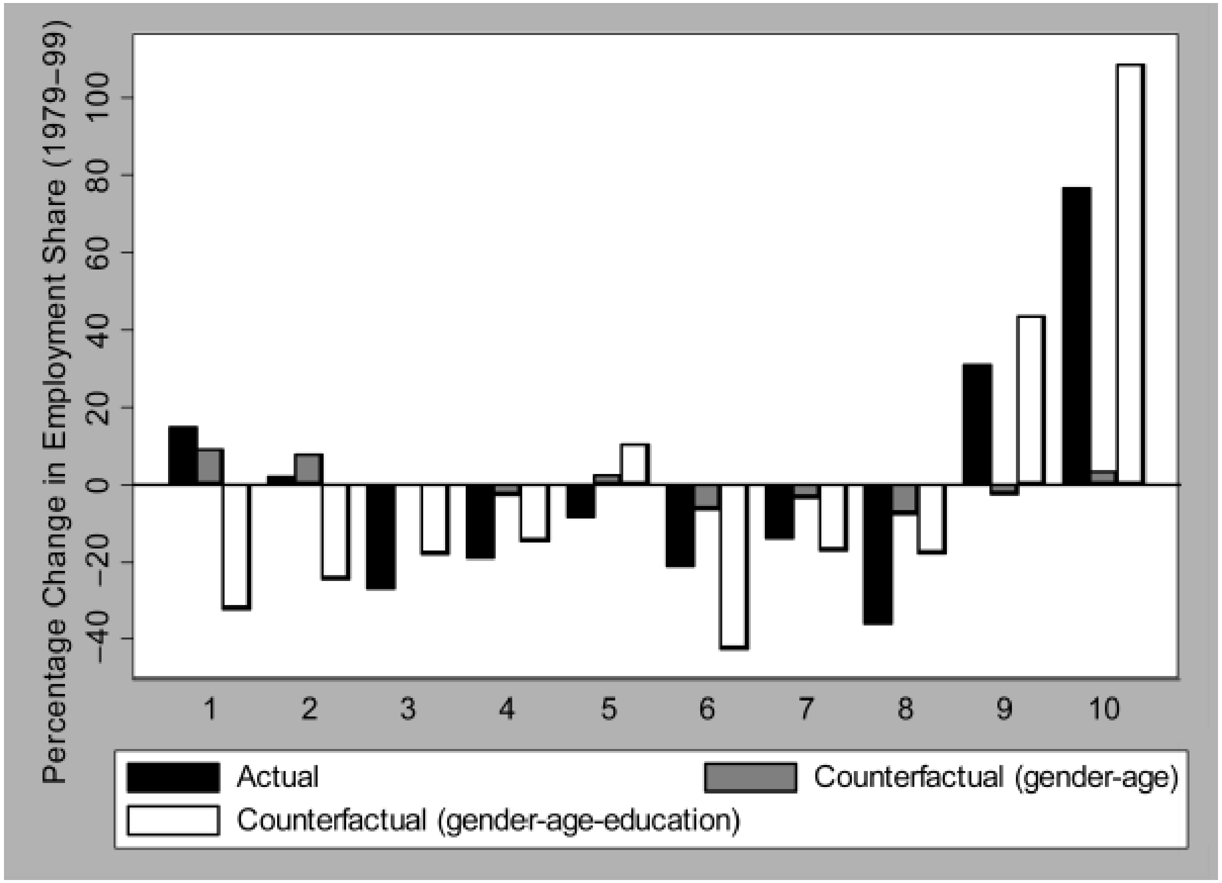
where Δn_j is the change in log employment in job j, and w_{j0} is the initial log median wage in the job.

TABLE 2A.—THE RELATIONSHIP BETWEEN EMPLOYMENT GROWTH AND INITIAL MEDIAN WAGE: MEN AND WOMEN TOGETHER

Sample	Sample Period	Data	Employment Measure	β_1	β_2	Fraction in Declining Section
Men + Women	1979–99	LFS (occ)	Employment	−4.541 (0.700)	2.107 (0.297)	52.93
Men + Women	1976–95	NES (occ)	Employment	−3.412 (0.664)	1.373 (0.267)	72.57
Men + Women	1979–99	LFS (occXind)	Employment	−4.804 (0.472)	2.109 (0.198)	62.80
Men + Women	1976–95	NES (occXind)	Employment	−3.957 (0.378)	1.581 (0.151)	74.69
Men + Women	1979–99	LFS (occ)	Hours	−4.218 (0.785)	2.047 (0.327)	28.42
Men + Women	1976–95	NES (occ)	Hours	−3.603 (0.775)	1.576 (0.319)	56.85
Men + Women	1979–99	LFS (occXind)	Hours	−4.331 (0.514)	1.969 (0.213)	49.67
Men + Women	1976–95	NES (occXind)	Hours	−4.145 (0.435)	1.748 (0.178)	62.22

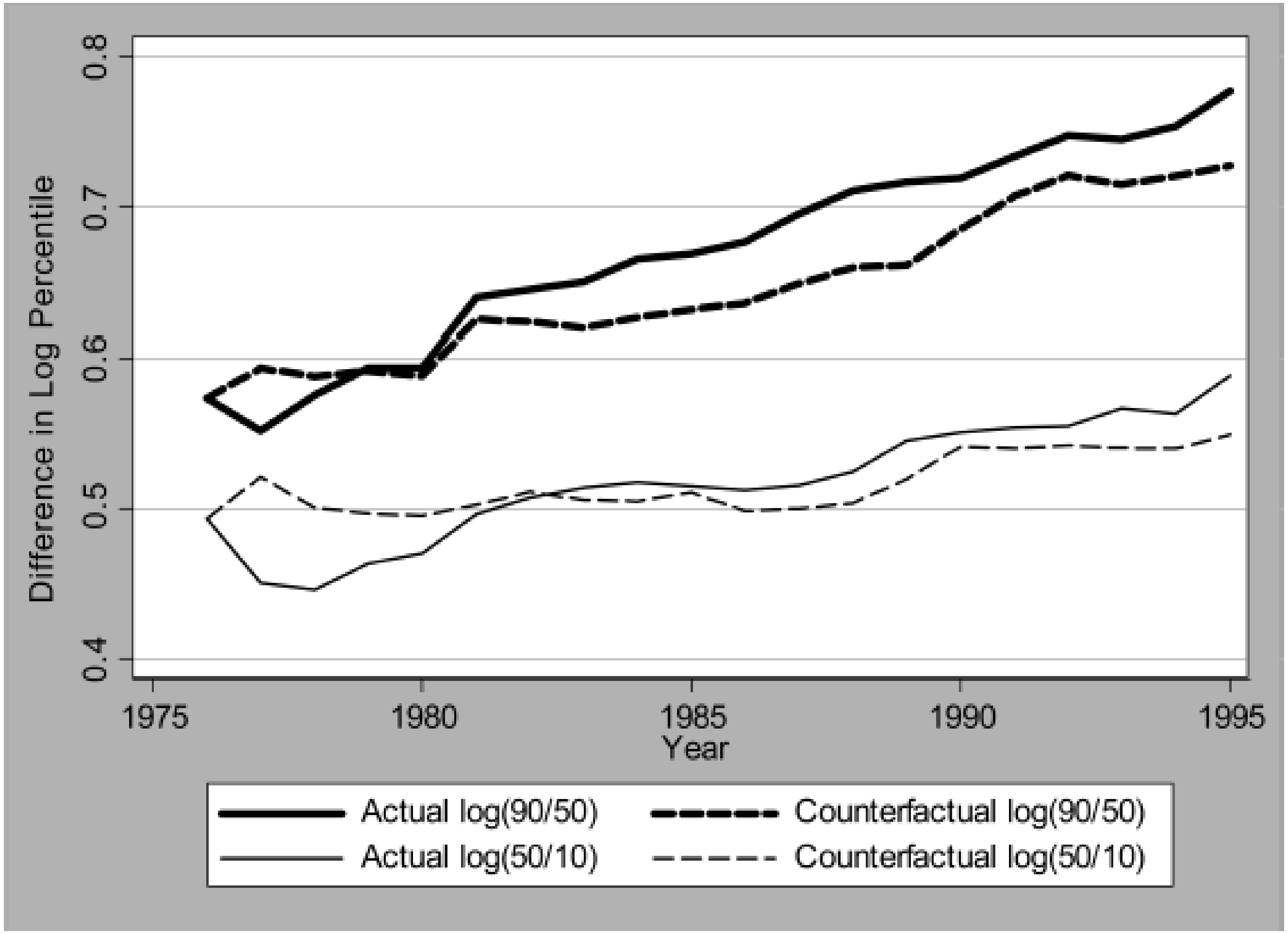
Notes: Regressions are weighted by job cell size in the initial period. Occupation uses three-digit SOC90 codes. Industry uses one-digit SIC80 codes.

FIGURE 4.—THE IMPACT OF CHANGES IN THE COMPOSITION OF THE LABOR FORCE



Notes: Employment data are taken from the LFS using sing three-digit SOC90 codes. Employment changes are taken between 1979 and 1999. Quality deciles are based on three-digit SOC90 median wages in 1979 taken from the NES. The first bar gives actual employment changes as in figure 1. The second bar gives counterfactual employment changes keeping the three-digit occupational composition within 24 gender-age cells constant over time. The third bar gives counterfactual employment changes keeping the three-digit occupational composition within 96 gender-age-education cells constant over time.

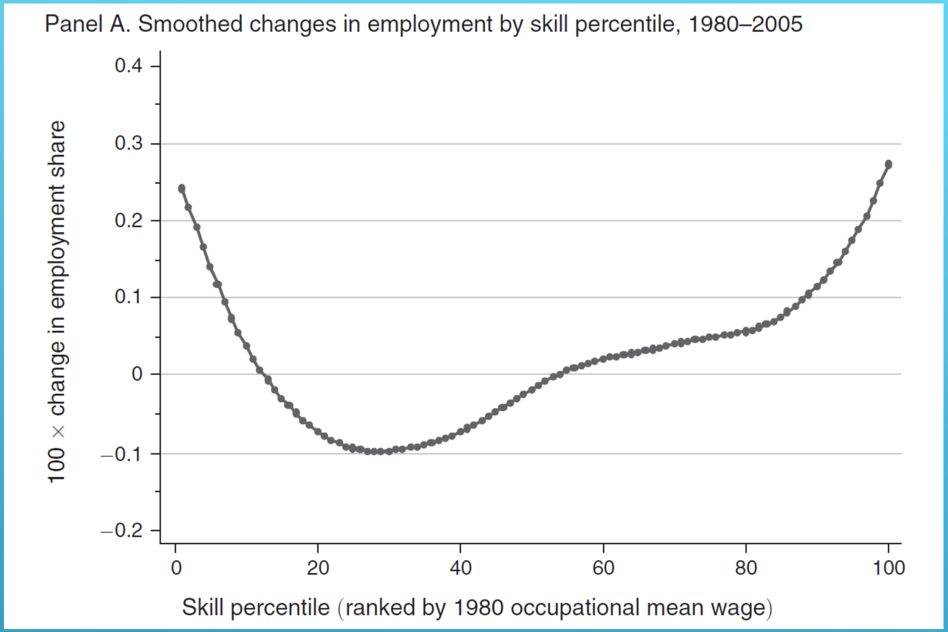
Figure 7.—The Impact of Job Polarization and Changing Relative Wages across Jobs on Wage Inequality

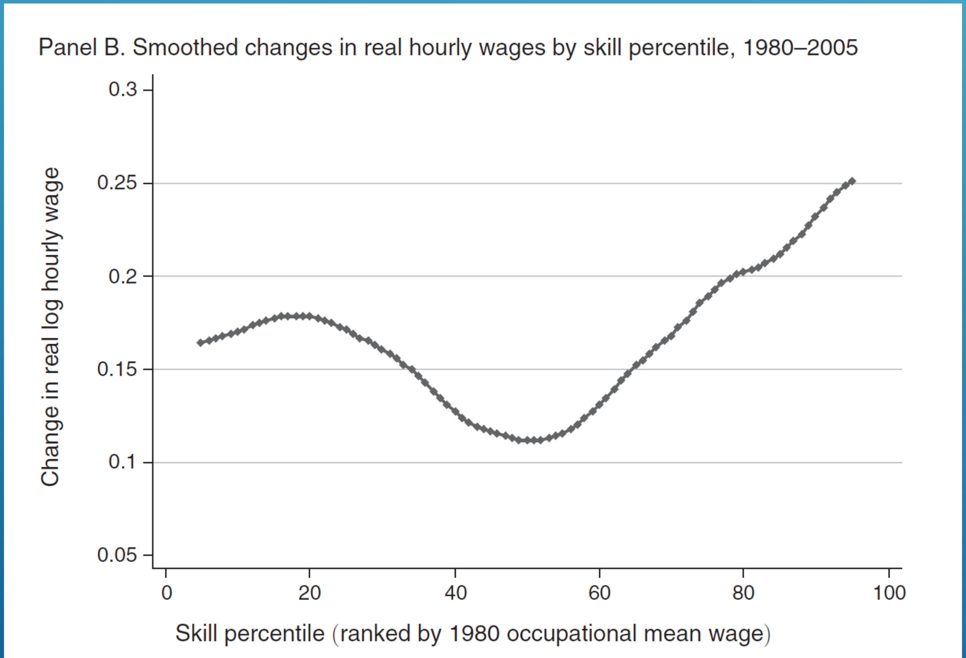


Notes: Data are taken from the NES. The figure uses three-digit SOC90 codes as the definition of a job. The counterfactual keeps constant wage dispersion within occupations but allows the actual median wage to vary in line with the data.

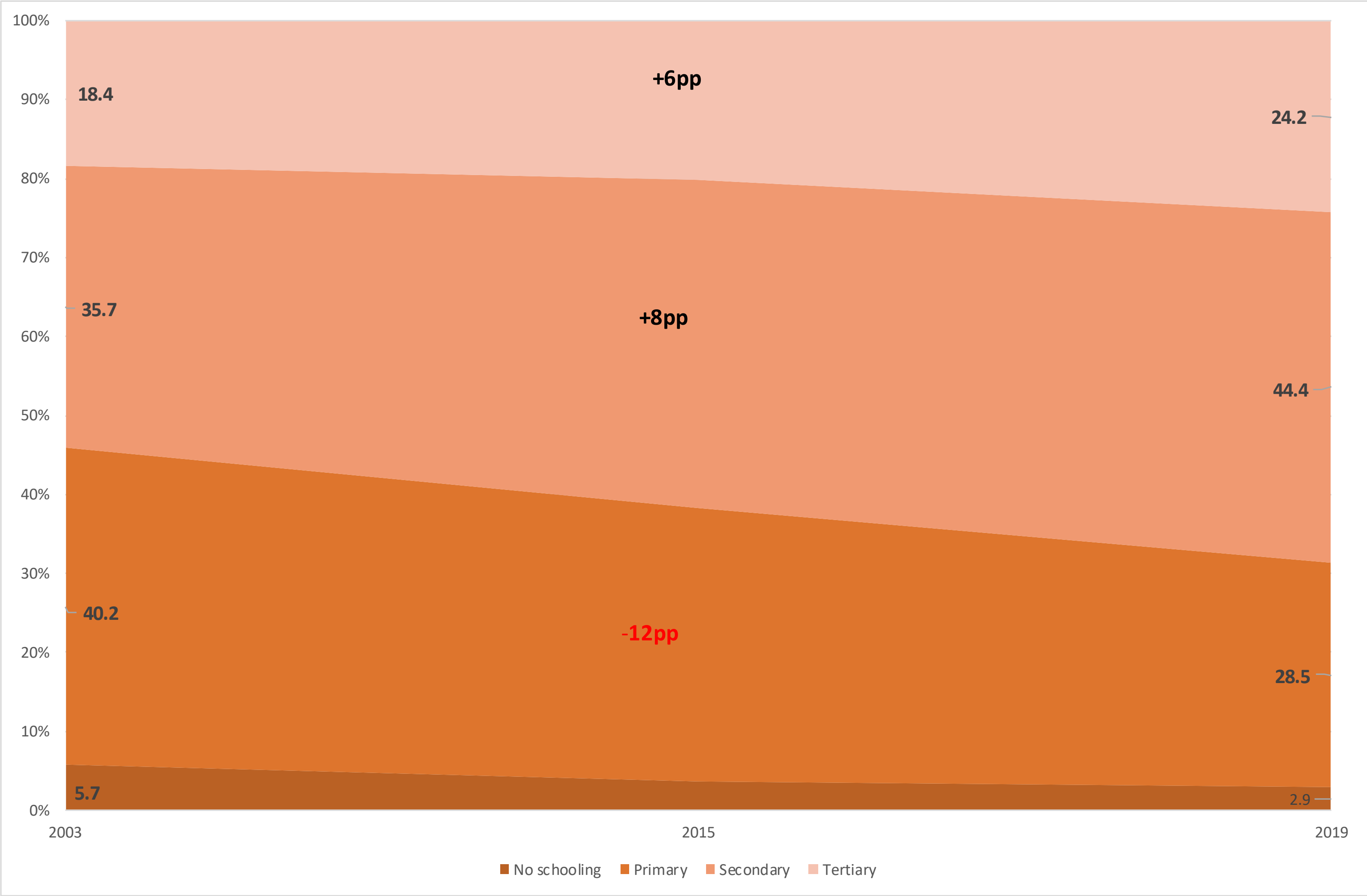
The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market†

By David H. Autor and David Dorn*

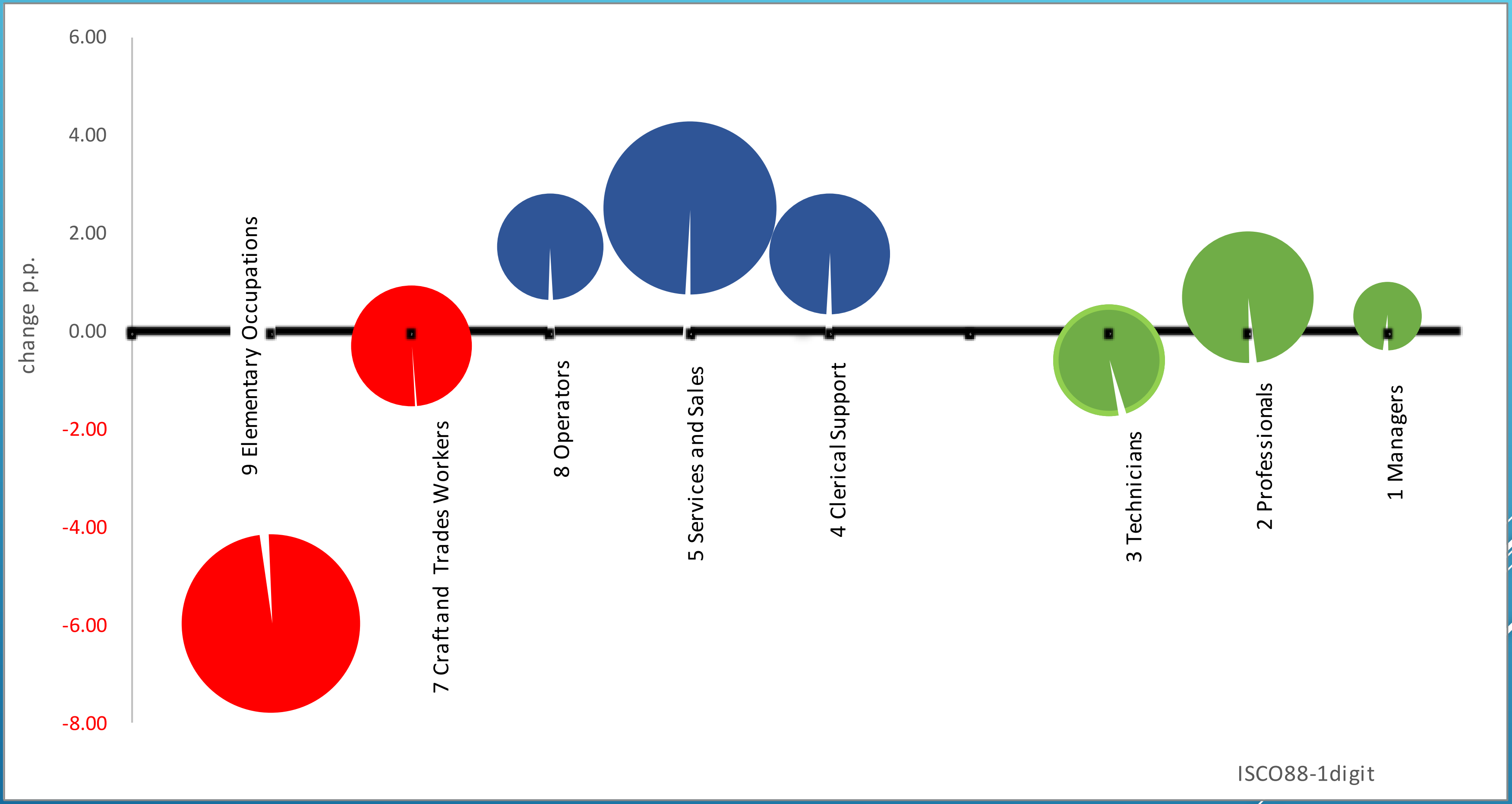

Panel A. Smoothed changes in employment by skill percentile, 1980–2005
100 × change in employment share
0.4
0.3
0.2
0.1
0
–0.1
–0.2
0
20
40
60
80
100
Skill percentile (ranked by 1980 occupational mean wage)


Panel B. Smoothed changes in real hourly wages by skill percentile, 1980–2005
Change in real log hourly wage
0.3
0.25
0.2
0.15
0.1
0.05
0
20
40
60
80
100
Skill percentile (ranked by 1980 occupational mean wage)

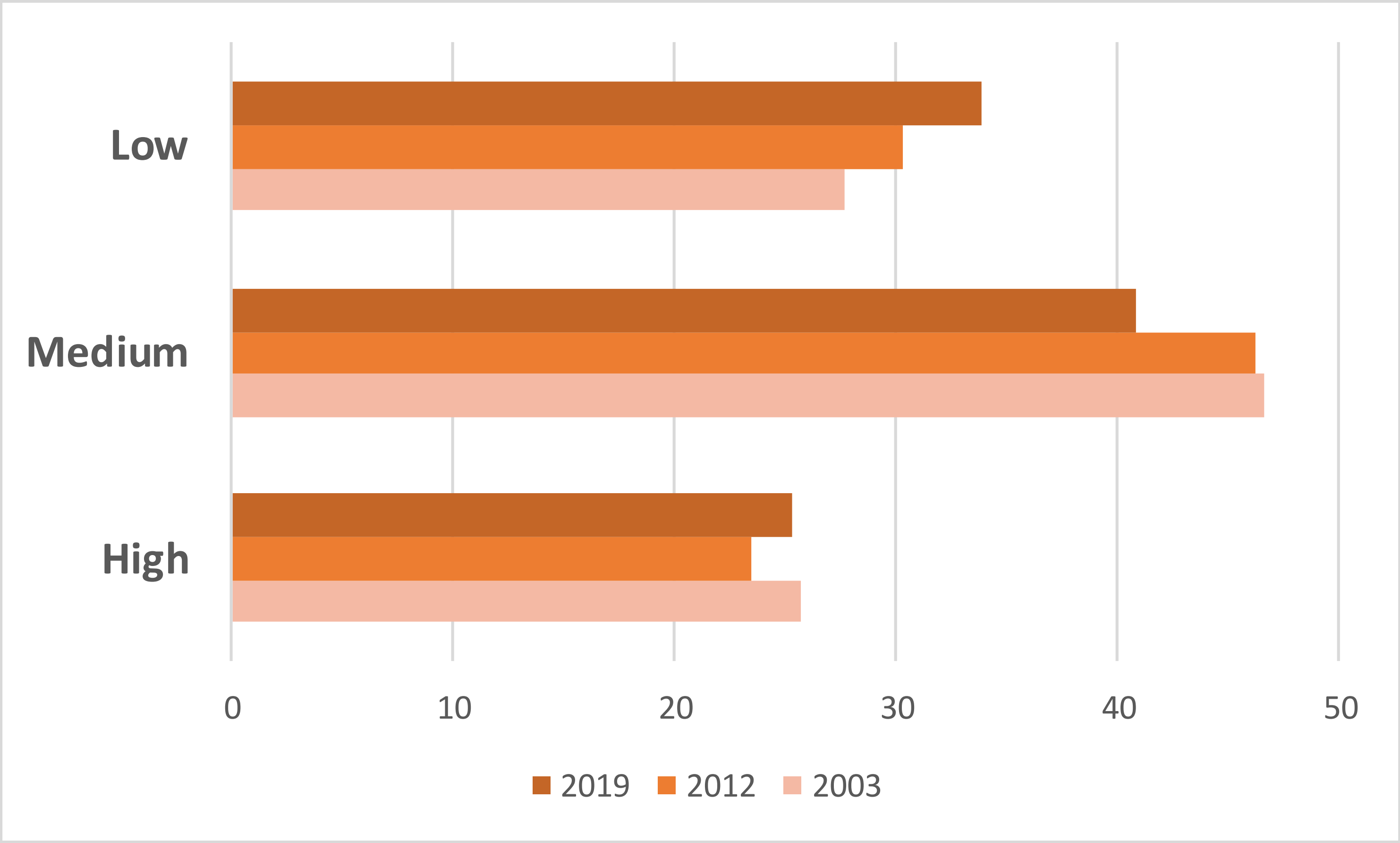
ARGENTINA 2003-2019



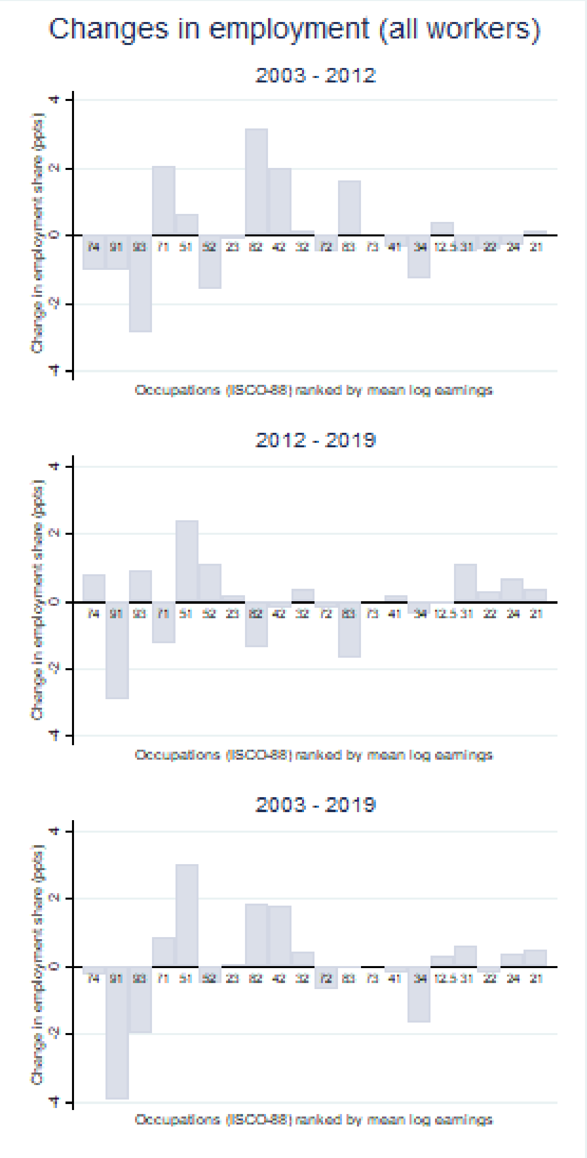
- A sharp decline in the % of elementary occupations. Increase in services and sale workers, plan and machine operators, and clerical support workers.

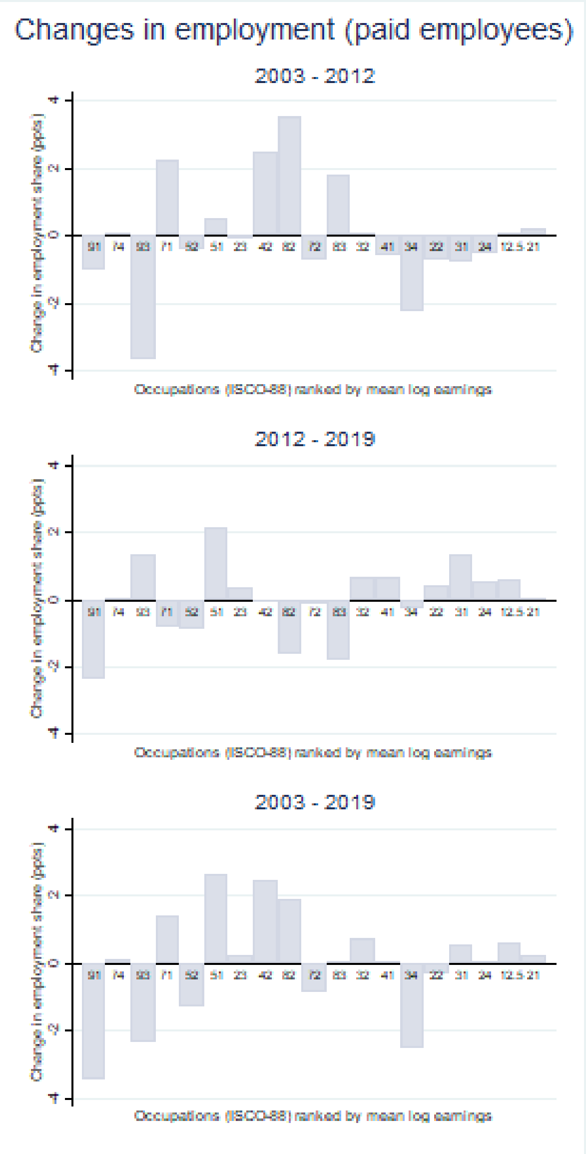


- Reduction of low-skilled jobs and increase in middle-skilled jobs. High-skilled jobs remained fairly stable over time.



- These changes resulted in a relocation of workers from low-paid to middle-paid jobs. High-skilled jobs remained fairly stable or slightly increased over time.



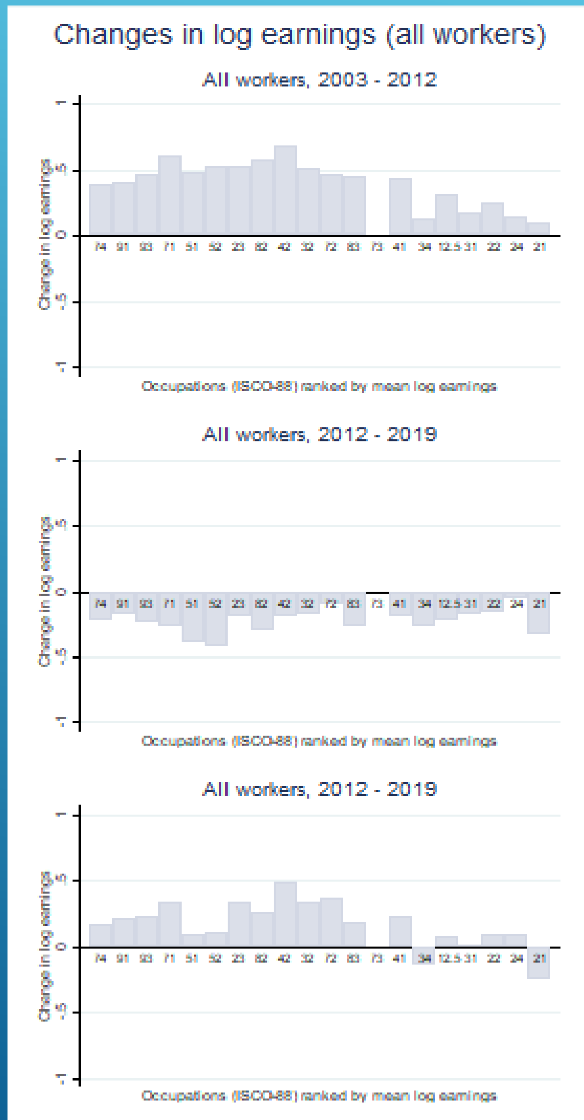


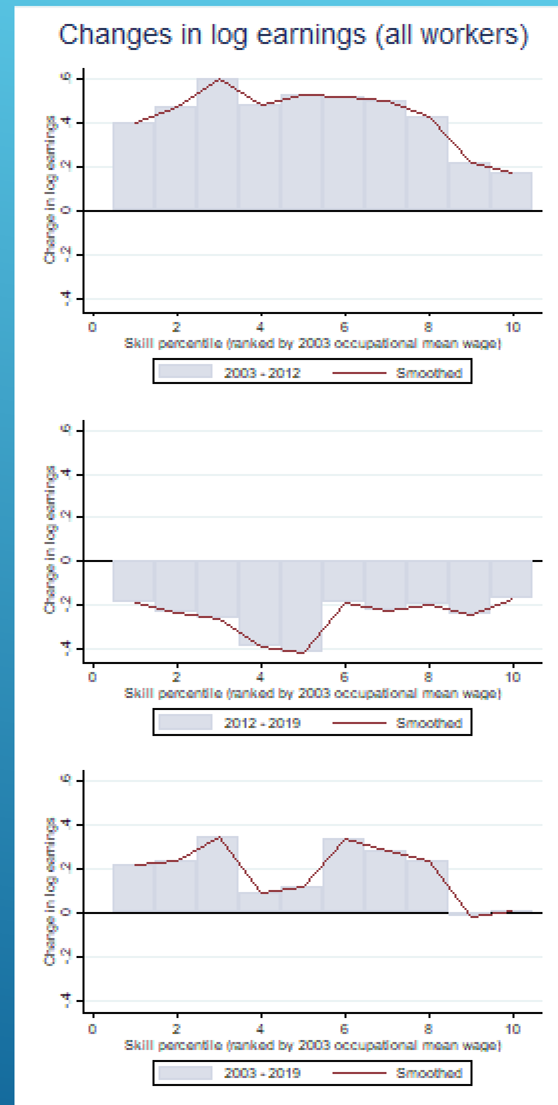
- $\Delta \log E_{i,t} = \beta_0 + \beta_1 \log(w_{j,t-1}) + \beta_2 \log(w_{j,t-1})^2$

	Log change in employment share					
	All workers			Paid employees		
Covariates	2003 - 2012	2012 - 2019	2003 - 2019	2003 - 2012	2012 - 2019	2003 - 2019
(log) mean hourly wage (t-1)	5.360	2.313	5.483	5.386	-1.499	4.681
	(3.896)	(4.758)	(3.206)	(3.386)	(3.823)	(3.043)
Sq. (log) mean hourly wage (t-1)	-0.332	-0.134	-0.334	-0.339	0.099	-0.284
	(0.243)	(0.285)	(0.203)	(0.214)	(0.231)	(0.194)
Constant	-21.625	-9.970	-22.460*	-21.395	5.587	-19.287
	(15.552)	(19.780)	(12.586)	(13.304)	(15.734)	(11.882)
Observations	20	20	20	19	19	19
R-squared	0.057	0.046	0.126	0.073	0.092	0.098
Adj. R-squared	-0.0538	-0.0665	0.0237	-0.0426	-0.0214	-0.0149
F test	0.383	0.707	0.0293	0.296	0.256	0.0384

Standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

- **First period**: real earnings growth was more intense in the middle part of the distribution. **Second period**: in a context of generalised fall of real earnings, the greatest reductions among middle-paid jobs.





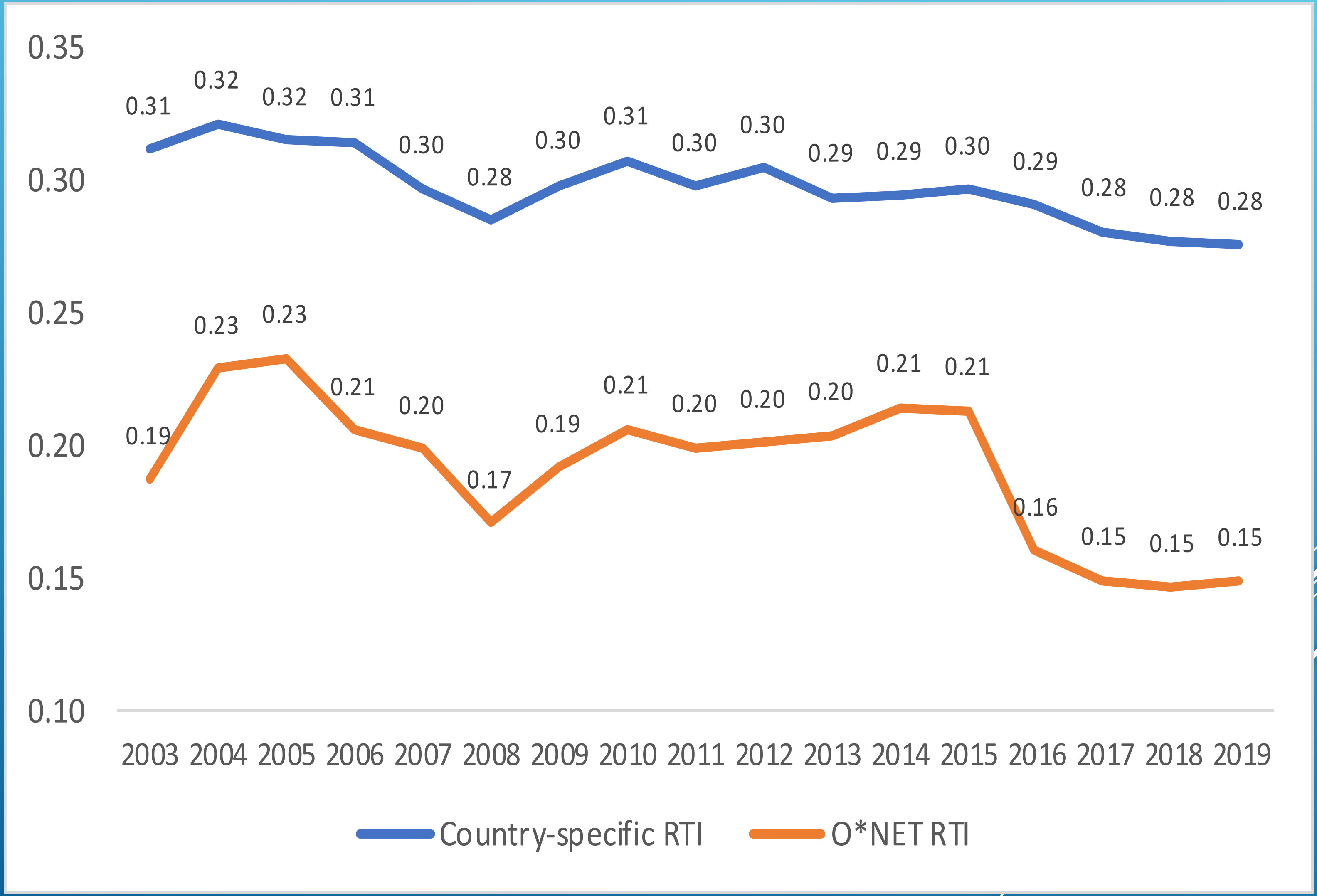
- First period: inverted U-shaped growth. Secod period: polarization pattern.

	All workers			Paid employees		
	Change in (log) mean wage			Change in (log) mean wage		
VARIABLES	2003 - 2012	2012 - 2019	2003 -2019	2003 - 2012	2012 - 2019	2003 -2019
(log) mean hourly wage (t-1)	6.703***	-5.773**	3.668**	5.848***	-3.480*	3.448*
	(0.765)	(2.263)	(1.675)	(1.190)	(1.730)	(1.681)
Sq. (log) mean hourly wage (t-1)	-0.429***	0.349**	-0.237**	-0.370***	0.210*	-0.219*
	(0.049)	(0.138)	(0.106)	(0.075)	(0.105)	(0.106)
Constant	-25.666***	23.553**	-13.941**	-22.613***	14.175*	-13.298*
	(2.962)	(9.265)	(6.574)	(4.676)	(7.098)	(6.654)
Observations	20	20	20	19	19	19
R-squared	0.750	0.362	0.314	0.556	0.268	0.198
Adj. R-squared	0.721	0.287	0.234	0.500	0.176	0.098
F test	0.000	0.036	0.043	0.001	0.089	0.068

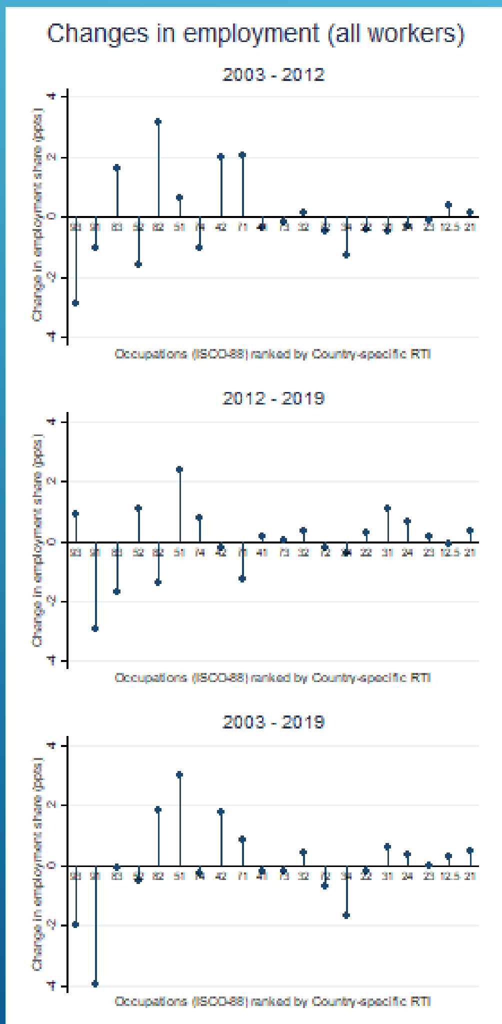
Standard errors in parentheses

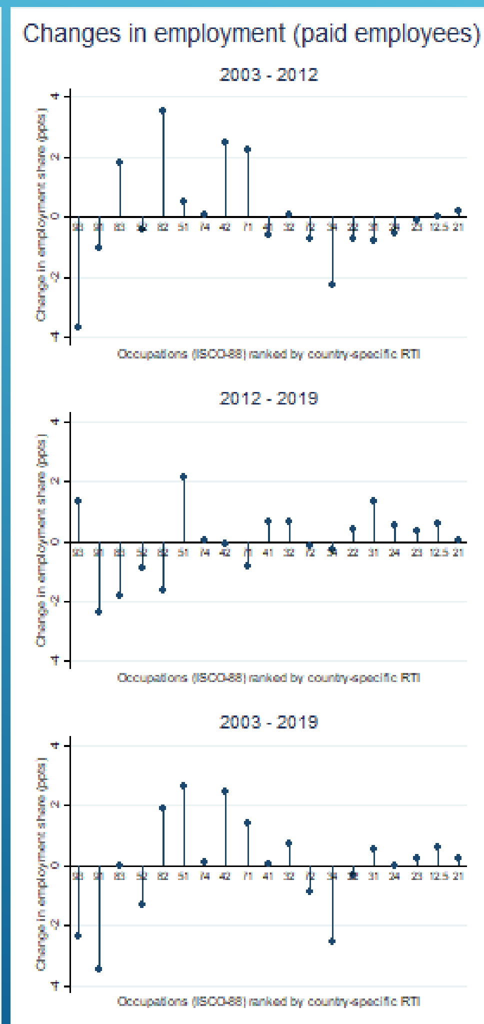
*** p<0.01, ** p<0.05, * p<0.1

- Decline in average RTI regardless of the measure used. It is, partly, associated with the reduction in the share of elementary occupations, which exhibit the highest level of routine tasks.



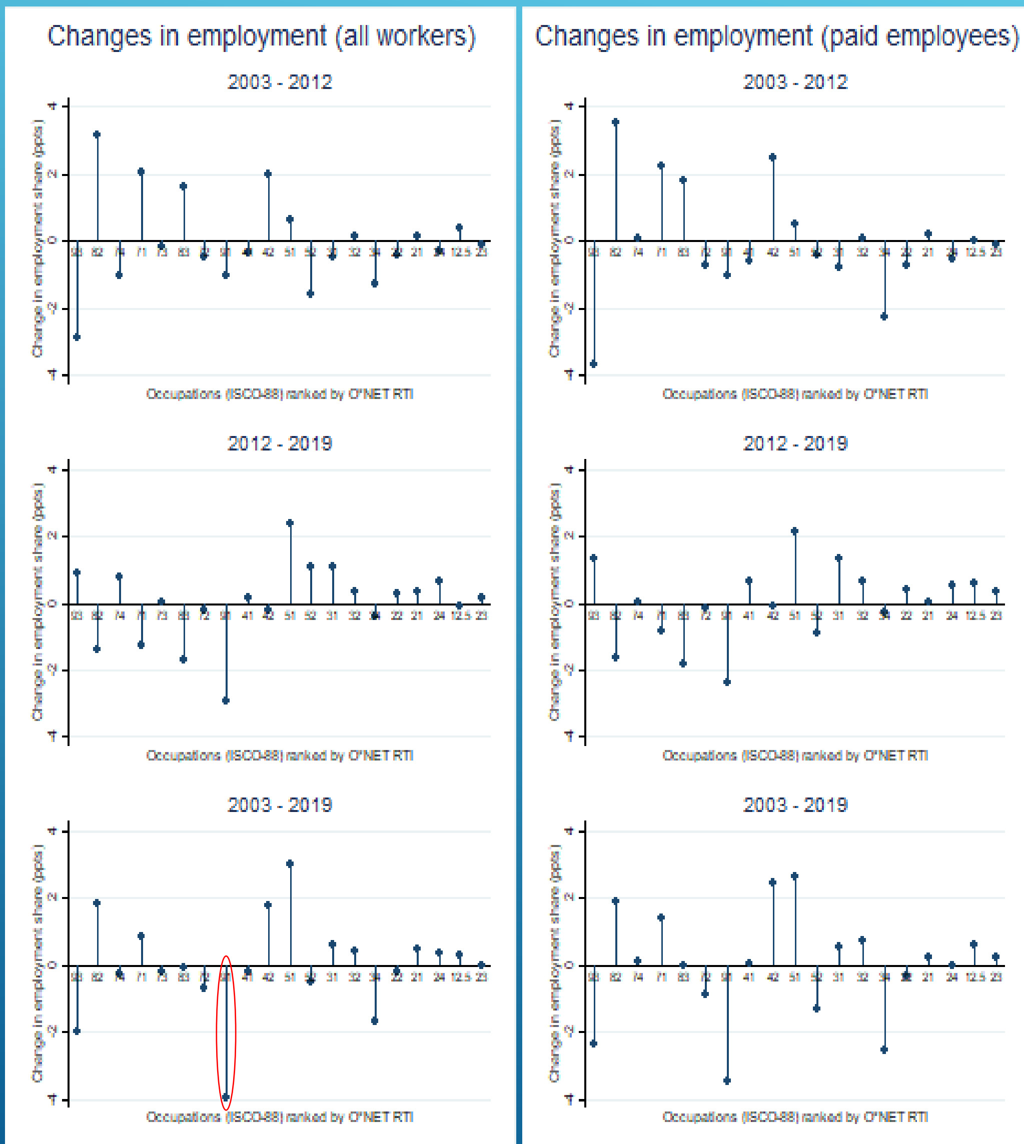
- **Country-specific RTI**: A loss of the share of occupations with high RTI, and less intense reduction (or some slight increases) with low RTI. Similar to the profile when jobs are ranked by mean earnings.
- The signs are consistent with this inverted U-shaped picture in the first period; however, **coefficients were not statistically significant**.





	Log change in employment share		
	2003 - 2012	2012 - 2019	2003-2019
Country specific RTI (t-1)	0.026	-0.104	-0.044
	(0.097)	(0.076)	(0.083)
Country specific RTI (t-1)_sq	-0.211	-0.085	-0.323*
	(0.221)	(0.166)	(0.161)

- **O*NET RTI**. The picture is very different.
- The coefficients were not statistically significant.



	Log change in employment share		
	2003 - 2012	2012 - 2019	2003-2019
O*NET RTI (t-1)	-0.026	-0.063	-0.053
	(0.067)	(0.043)	(0.044)
O*NET RTI (t-1)_sq	-0.023	-0.013	-0.014
	(0.048)	(0.034)	(0.033)

Changes in the employment share and earnings according to Europe's job ranking